

Note 4.4 Concavity and Curve Sketching

Concavity

- **Definition:**

If the function $f(x)$ is differentiable on (a, b) then the graph of f is concave upward(downward) on (a, b) if f' is increasing(decreasing).

- **The Second Derivative Test For Concavity:**

Let $y = f(x)$ is twice-differentiable on I , then the function f is concave upward(downward) when $f''(x) > 0(f''(x) < 0)$ on I .

- **The Procedure of Finding Concave Intervals**

- Finding the points where $f''(x) = 0$ or $f''(x)$ doesn't exist.
- Make the sign table with intervals, $f'(x), f''(x)(CU(\cup), CD(\cap))$
- Write the answer.

Points of Inflection

- **Definition:**

A point where the graph of a function has a tangent line and the concavity changes. At such a point, $f''(c) = 0$ or $f''(x)$ doesn't exist.

Second Derivative Test for Local Extrema

- **Theorem 5--Second Derivative Test for local Extrema:**

Suppose f'' is continuous on an open interval that contains $x = c$ and $f'(c) = 0$.

- If $f''(c) > 0(f''(c) < 0)$, then f exists a local minimum(maximum) at $x = c$.
- If $f''(c) = 0$ or doesn't exist, the test is inconclusive.

- **Procedure for Graphing $y = f(x)$**

- Find the intercepts of $f(x)$: Solve $f(x) = 0$ for x and substitution $x = 0$ to get the x intercept(s) and the y intercept.
- Find $f'(x), f''(x)$: Solve $f'(x) = 0, f''(x) = 0$ for x .
- Make a sign table with interval, $f(x), f'(x), f''(x)$.
- Sketch the graph of f .

- **Guidelines For Sketching a Curve:**

- Domain
- Intercepts
- Symmetry
- Asymptotes
- Intervals of Increase and decrease

- Concavity and points of inflection
- Sketch the curve
- **Strategy for graphing a rational function:**
 - Domain
 - Intercepts
 - Vertical and horizontal asymptotes
 - Find $f'(x)$, $f''(x)$ and where $f'(x)$, $f''(x) = 0$ or don't exist.